

Problem 1

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- (a) True or False: To find $\lim_{x \rightarrow 2} f(x)$, it's enough to know the values of $f(2.1)$, $f(2.01)$, $f(2.001)$, and so on.

Explanation. False. These values will only help us make a guess at $\lim_{x \rightarrow 2^+} f(x)$, the right hand limit of f as x approaches 2. To determine $\lim_{x \rightarrow 2} f(x)$, we also need to know $\lim_{x \rightarrow 2^-} f(x)$ which we cannot determine from the above values. For example, consider the function

$$f(x) = \begin{cases} -1 & \text{if } x < 2, \\ 0 & \text{if } x = 2, \text{ and} \\ 1 & \text{if } 2 < x. \end{cases}$$

Looking at the graph of this function below, we can see that $\lim_{x \rightarrow 2^+} f(x) = 1$ and $\lim_{x \rightarrow 2^-} f(x) = -1$. Thus, since $\lim_{x \rightarrow 2^+} f(x) \neq \lim_{x \rightarrow 2^-} f(x)$, $\lim_{x \rightarrow 2} f(x)$ does not exist.

alt=graph of function that is 1 for $x > 2$, -1 for $x < 2$, and with $f(2)=0$

- (b) True or False: If we know the value of $f(2)$, then can we conclude that $\lim_{x \rightarrow 2} f(x) = f(2)$?

Explanation. False. In the above example, we have that $f(2) = 0$, while $\lim_{x \rightarrow 2} f(x)$ does not exist.